

* Topology Recap.

Main referene: Enjoying math.

def. (Topology)

Given a set $X (\neq \emptyset)$, we say $\mathcal{T} \subseteq \mathcal{P}(X)$ is a topology on X .

if $\mathcal{T}$ satisfies the following ①, ②, ③:

① $\emptyset, X \in \mathcal{T}$

② If $\{E_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}$, then $\bigcup_{\alpha \in I} E_\alpha \in \mathcal{T}$.

→ "임의의" index set I 에 대해 성립.

③ If $\{E_k\}_{k \in \mathbb{N}} \subseteq \mathcal{T}$, then $\bigcap_{k \in \mathbb{N}} E_k \in \mathcal{T}$.

* Topology를 정의한다 / 부여한다.

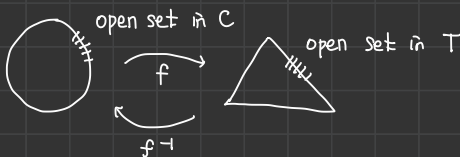
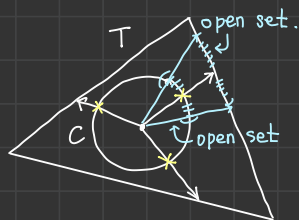
⇔ 공간 내의 "open set"이 무엇인지 규정한다.

We will call the element of $\mathcal{T}$ the open set.

Rmk. (연속성과 Topology의 관계.)

$$f: C \rightarrow T$$

$$f^{-1}: T \rightarrow C$$



We will call f (which maps the open set in C to the open set in T) the continuous function.

def. (Topological space)

If a set X is equipped with a topology $\mathcal{T}$, we will call $(X, \mathcal{T})$ a topological space.

Ex. (i) $(X, \mathcal{P}(X))$ is topological space. (every subset of X is open set.)

: called the discrete topological space. (각 원소 $\{x_1\}, \{x_2\}, \dots$ 이렇게 존재할 수 있어서 discrete.)

(ii) $(X, \mathcal{T})$, $\mathcal{T} = \{\emptyset, X\}$ (the most minimal topological space.)

: called the indiscrete topological space. (항상 묶여다녀야해서 indiscrete.)

def. Given a set X , let $\mathcal{T}, \mathcal{T}'$ be topology on X .

We say $\mathcal{T}'$ is finer (larger) than $\mathcal{T}$ if $\mathcal{T} \subseteq \mathcal{T}'$.

$\mathcal{T}'$ is coarser (smaller) than $\mathcal{T}$ if $\mathcal{T} \supseteq \mathcal{T}'$.

In classical analysis, we use the metric topology.

def. (metric)

Given a set X , $d: X \times X \rightarrow \mathbb{R}$ is a metric if

① $\forall x, y \in X, d(x, y) \geq 0$ and $d(x, y) = 0$ iff $x = y$

② $\forall x, y \in X, d(x, y) = d(y, x)$

③ $\forall x, y, z \in X, d(x, y) \leq d(x, z) + d(z, y)$

def. (open set on metric space.)

Given (X, d)

① Given $\varepsilon > 0, p \in X, N_{\varepsilon}(p) = \{x \in X : d(x, p) < \varepsilon\}$ called the ε -neighborhood of p .

② We say that U is open in X if

$$\forall p \in U, \exists \varepsilon_p > 0 \text{ s.t. } N_{\varepsilon_p}(p) \subseteq U.$$

Thm. Given (X, d) , define $\mathcal{T}_d = \{U \subseteq X : \forall p \in U, \exists \varepsilon_p > 0 \text{ s.t. } N_{\varepsilon_p}(p) \subseteq U\}$

Then, $\mathcal{T}_d$ is topology on X .

: called the metric topology. (generated by given metric "d")

: So, it is reasonable to say that U is open.

Rmk. $\lim_{n \rightarrow \infty} a_n = a$

$$\iff \forall \varepsilon > 0, \exists N_{\varepsilon} \in \mathbb{N} \text{ s.t. } n \geq N_{\varepsilon} \Rightarrow d(a_n, a) < \varepsilon$$

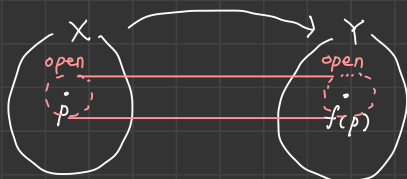
$$\iff \forall \varepsilon > 0, \exists N_{\varepsilon} \in \mathbb{N} \text{ s.t. } n \geq N_{\varepsilon} \Rightarrow a_n \in N_{\varepsilon}(a)$$

def. (continuity of function.)

A function $f: E(\subseteq \mathbb{R}) \rightarrow \mathbb{R}$ is continuous at $p \in E$ if,

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } |x - p| < \delta \Rightarrow |f(x) - f(p)| < \varepsilon$$

$$\iff \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } x \in N_{\delta}(p) \Rightarrow f(x) \in N_{\varepsilon}(f(p))$$



def. (Topological view of sequence.)

Given a topological space $(X, \mathcal{T})$,

① A sequence $f: \mathbb{N} \rightarrow X$ is a function from $\mathbb{N}$ to X .
$$n \mapsto a_n$$

② Let $\alpha \in X$, $(a_n)_{n=1}^{\infty} \subseteq X$ be sequence in X .

We say $(a_n)_{n=1}^{\infty}$ converges to α if

$\forall$ open set V containing α , $\exists N = N(V) \in \mathbb{N}$ s.t. $n \geq N \Rightarrow a_n \in V$.

In this case, denote $\lim_{n \rightarrow \infty} a_n = \alpha$.

Rmk. open set V containing $\alpha \equiv V$ is the neighborhood of α .

Rmk. 수열의 수렴성은 위상공간에서 어떤 위상을 선택하는지에 따라 달라진다.

$X = \mathbb{N}$, $\mathcal{T} = \mathcal{T}_c = \{U \subseteq \mathbb{N} : U^c \text{ is finite}\} \cup \{\emptyset\}$

Define $a_n := n \quad \forall n \in \mathbb{N}$.

Claim: a_n converges to every points in $\mathbb{N}$.

Take any $m \in \mathbb{N}$.

Take any neighborhood V of m .

Then, V^c must be finite. Thus we can define $N = \max V^c + 1$.

Then, $\forall n \geq N$, $n \notin V^c$. i.e. $n \in V$. ▨

open set에 따라 비직관적인 결과가 도출할 수 있다.

위 상황을 피하기 위해 "좋은" 위상공간을 고르려고 노력한다. 그럼 좋은게 뭐지?

→ "Hausdorff axiom."

Hausdorff axiom을 만족하는 위상공간 위의 수열은 수렴의 유일성이 보장됨으로 알려져 있다.

(위에서는 모든 점으로 수렴.)

def. (basis) $\leftarrow$ Linear Algebra의 basis와는 다른 대상이다. 근데 비슷한 감성이다.

If $\mathcal{B} \subseteq \mathcal{P}(X)$ satisfies

① $\forall x \in X, \exists B \in \mathcal{B} \text{ s.t. } x \in B$

② If $x \in B_1 \cap B_2, x \in X, B_1, B_2 \in \mathcal{B}$, then $\exists B_3 \in \mathcal{B} \text{ s.t. } x \in B_3 \subseteq B_1 \cap B_2$

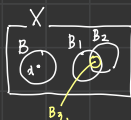
We call such $\mathcal{B}$ the basis on X & call each element of $\mathcal{B}$ a basis element.

e.g. (plane)

$X = \text{plane}$.

$\mathcal{B}_1 = \{ \text{circles in plane} \}$

$\mathcal{B}_2 = \{ \text{rectangles in plane} \}$



def. (open)

Given a set X , $\mathcal{B}$ be a basis of X .

We say U is open in X if

$$\forall x \in U, \exists B_x \in \mathcal{B} \text{ s.t. } x \in B_x \subseteq U. \dots (*)$$

(i.e. $U = \bigcup_{x \in U} B_x$)

Thm.

Given a set X and corresponding basis $\mathcal{B}$.

Then, $\gamma = \{ U \subseteq X : U \text{ is open in the sense of } (*) \}$ is a topology on X .

pf. ① check $\emptyset, X \in \gamma$

Let $x \in X$.

Since $\mathcal{B}$ is a basis, $\exists B_x \in \mathcal{B} \text{ s.t. } x \in B_x \subseteq X$.

Thus, X is open in the sense of $(*)$.

$\emptyset \in \gamma$ is vacuously true.

② check $\bigcup_{\alpha \in I} U_\alpha \in \gamma$

Let $x \in \bigcup_{\alpha \in I} U_\alpha$.

Then $\exists \alpha_0 \in I \text{ s.t. } x \in U_{\alpha_0}$.

But, since $U_{\alpha_0} \in \gamma$, $\exists B_x \in \mathcal{B} \text{ s.t. } x \in B_x \subseteq U_{\alpha_0}$.

Thus, $B_x \subseteq \bigcup_{\alpha \in I} U_\alpha$.

③ check $\bigcap_{i=1}^n U_i \in \gamma$

Let $U, V \in \gamma$ and $x \in U \cap V$.

Then $x \in U$ and $x \in V$.

Observe that $\exists B_U \in \mathcal{B} \text{ s.t. } x \in B_U \subseteq U$ and $\exists B_V \in \mathcal{B} \text{ s.t. } x \in B_V \subseteq V$.

Thus, $x \in B_u \cap B_v$.

Since $B_u, B_v \in \mathcal{B}$, $\exists B_x \in \mathcal{B}$ s.t. $x \in B_x \subseteq B_u \cap B_v$. ($\because$ def of Basis.)

Therefore, $x \in B_x \subseteq U \cap V$.

eg. (Metric topology. (X, d))

open ball $B_r(x) := \{y \in X : d(x, y) < r\}$: basis element.

$\rightarrow U = \bigcup_{x \in U} B_r(x)$: open set.

$\rightarrow \mathcal{T} = \{\text{such } U\}$: metric topology. (why?)

Prop. (더 작게 주어진 Basis도 해당하는 위상도.)

Given a set X . Let $\mathcal{B}, \mathcal{B}'$ be basis on X .

($\rightarrow \mathcal{T}$) ($\rightarrow \mathcal{T}'$)

Then $\mathcal{T} \subseteq \mathcal{T}'$ if and only if $\forall B \in \mathcal{B} \forall x \in B \exists B'_x \in \mathcal{B}'$ s.t. $x \in B'_x \subseteq B$.

pf. ($\Rightarrow$)

Let $\mathcal{T} \subseteq \mathcal{T}'$.

choose any $B \in \mathcal{B}$, take any $x \in B$.

Since $x \in B$, $B \in \mathcal{T} \subseteq \mathcal{T}'$.

$$\Rightarrow B = \bigcup_{x \in B} B'_x$$

$$\Rightarrow \exists x_0 \in B'_{x_0} \text{ s.t. } x_0 \in B'_{x_0} \subseteq B.$$

($\Leftarrow$)

To show $\mathcal{T} \subseteq \mathcal{T}'$, take any $U \in \mathcal{T}$.

claim: $U \in \mathcal{T}'$

$$\text{Write } U = \bigcup_{x \in U} B_x.$$

Since $\forall x \in U$, $B_x \in \bigcup_{y \in B_x} B'_y$ by assumption.

Thus, $U = \bigcup_{x \in U} \left(\bigcup_{y \in B_x} B'_y \right)$: union of basis element in $\mathcal{B}'$.

$\therefore U \in \mathcal{T}'$



* Measure Theory.

def. (σ -algebra.)

X : set

If $m \subseteq \mathcal{P}(X)$ satisfies

① $X \in m$

② If $A \in m$ then $A^c \in m$

③ If $\{A_n\}_{n=1}^{\infty} \subseteq m$ then $\bigcup_{n=1}^{\infty} A_n \in m$

* " σ " is term countable을 지칭하며 종종 사용된다.

* 통계학에서는 sample space의 σ -algebra를 고려하고,
그에 대한 measurable set을 "event"라고 지칭한다.

We call an element of σ -algebra a "measurable set."

Rem. (i) ③ 정리에서 countable을 finite로 제한하면 "algebra"가 된다.

(i.e. If $\{A_n\}_{n=1}^k \subseteq m$ then $\bigcup_{n=1}^k A_n \in m$.)

(ii) By ①, $X \in m$

By ②, $X^c = \emptyset \in m$.

Let $A, B \in m \Rightarrow A^c, B^c \in m$.

Then $A^c \cup B^c \cup \emptyset \cup \emptyset \cup \dots \cup \emptyset \in m$

$= A^c \cup B^c$

Again, $(A^c \cup B^c)^c = A \cap B \in m$. "closed under intersection!!"

Moreover, If $\{A_n\}_{n=1}^{\infty} \subseteq m$, then $\bigcup_{n=1}^{\infty} A_n \in m$.

If $A, B \in m$, then $A - B \in m$ ($A - B = A \cap B^c$)

(iii) In general, $\exists$ topology $\mathcal{T}$ on some set X s.t. $|\mathcal{T}| = |\mathcal{N}|$

However, $\nexists$ σ -algebra m of X such that m is countable.

(topology와 σ -algebra는 유사해보이나 다르다)

def. Given m σ -algebra of X and topological space $(Y, \mathcal{T})$,

we say that $f: X \rightarrow Y$ is a measurable function

if $\forall U \in \mathcal{T}, f^{-1}(U) \in m$.

Rmk. Let $X \xrightarrow[\text{measurable}]{f} Y \xrightarrow[\text{continuous}]{g} Z$ Continuous 와 measurable 함수의 합성은 measurable 이다. $\lceil$

$\searrow$
 $g \circ f$

By definition, $g \circ f$ is measurable. ($\because (g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)) \in m$)

Proposition.

Given measurable space (X, m) . topological space $(Y, \mathcal{T}_Y)$ and $u, v : X \rightarrow \mathbb{R}$ be measurable.

Let $\Phi : \mathbb{R} \times \mathbb{R} \rightarrow Y$ be continuous.

Define $h : X \rightarrow Y, x \mapsto \Phi(u(x), v(x))$

Then, h is measurable.

pf It suffices to show that $x \mapsto (u(x), v(x))$ measurable.

Let $(a_1, b_1) \times (a_2, b_2) \in \mathbb{R} \times \mathbb{R}$.

Then $f^{-1}((a_1, b_1) \times (a_2, b_2)) = \underbrace{u^{-1}(a_1, b_1)}_{\in m} \cap \underbrace{v^{-1}(a_2, b_2)}_{\in m} \in m$.

Rmk. Let $E \in m$. (i.e. E is a measurable set.)

Define $\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$

$\chi_E : X \rightarrow \mathbb{R}$ is measurable.

정리대로 확인하면 된다. $\forall B \subset \mathbb{R}, f^{-1}(B) \in m$ 인지 check.

$$\chi_E^{-1}(B) = \begin{cases} X & 0 \in B, 1 \in B \\ E & 0 \notin B, 1 \in B \\ E^c & 0 \in B, 1 \notin B \\ \emptyset & 0 \notin B, 1 \notin B. \end{cases}$$

하면, by definition, $E, E^c, \emptyset, X \in m$. ▣

prop. Let $\mathcal{F} \subseteq \mathcal{P}(X)$.

Then, $\exists$ smallest σ -algebra m^* of X which contains $\mathcal{F}$.

그냥 $\mathcal{F}$ 를 포함하는 σ -algebra는 trivial 하게 존재한다.

pf. Take $m^* := \bigcap_{\mathcal{F} \subset m} m$

Need to check (i) m^* is σ -algebra

(ii) m^* is smallest.

def. (Borel σ -algebra.)

Given (X, γ) topological space.

We call m^* the smallest σ -algebra containing γ

the Borel σ -algebra $\mathcal{B}$ of X . ($m^* = \bigcap_{\gamma \subseteq m} m$)

The element of Borel σ -algebra is called a Borel set.

Rmk. ① Every open set is a Borel set.

② Every closed set is a Borel set.

③ $\bigcap_{n=1}^{\infty} U_n$ & $\bigcap_{n=1}^{\infty} F_n$: Borel set. where U_n open, F_n closed.